

Education

IIT Bombay <i>M.Tech in Industrial Engineering and Operations Research (IEOR)</i>	Jul 2024 – Jul 2026
Guru Ghasidas University <i>B.Tech in Computer Science and Engineering (CSE)</i>	Jul 2020 – Jul 2024
• Relevant Coursework: Deep Learning, Reinforcement Learning, NLP, Generative & Agentic AI, ML Principles, Operations Research	

Experience

Full-Stack Developer Intern <i>Jumps Automation</i> Powai, Maharashtra	May 2026 – Present
• Developed CyberAegis , a gamified cybersecurity training platform with 7 microservices and 41 interactive simulators	
• Engineered 128+ REST API endpoints with an event-driven architecture using RabbitMQ and Redis, serving a React 18 SPA	
• Deployed 9 containerized services on Kubernetes; wrote 261+ integration tests validated for 5,000 concurrent users	
Full-Stack Developer Intern <i>Vajra Labs, IIT Bombay</i> Powai, Maharashtra	Aug 2025 – Dec 2025
• Built a malware analysis platform processing 200+ daily submissions with automated feature extraction pipelines	
• Engineered sandboxed multi-VM execution with an asynchronous Celery pipeline for high-throughput concurrent analysis	
Full Stack Intern <i>Dimensionless Technologies</i> Navi Mumbai, Maharashtra	Nov 2021 – Jan 2022
• Built an image-processing pipeline using OpenCV.js and Google Vision API to identify medicines from batch photos	
• Auto-populated an Excel-compatible inventory system , cutting manual medicine cataloging from hours to minutes per batch	

Technical Skills

• Languages (Proficient): Python, C, C++, JavaScript, TypeScript, SQL	• (Familiar): Go, Rust, Java, Bash
• ML & Data: PyTorch, HuggingFace Transformers, Scikit-Learn, NumPy, Pandas, Matplotlib, OpenCV, RLHF, LLM Agents	
• Frameworks: Django, DRF, FastAPI, React.js, Next.js, Node.js, Tailwind CSS, Jupyter, Streamlit	
• Systems & Tools: PostgreSQL, Neo4j, Redis, Docker, Docker Compose, Git, GitHub Actions, Ray, Celery, RabbitMQ, Linux	

Projects

Explainable Depression Detection from Social Media <i>Course Project</i> Prof. Pushpak Bhattacharyya	Aug 2025 – Nov 2025
• Built an NLP pipeline for depression detection from social media posts, using double domain adaptation and guided masking	
• Designed an explainability pipeline by combining Integrated Gradients with an LLM-based teacher-student distillation method	
• Achieved an F1 score of 0.78 on test data, and extracted top-k words contributing most strongly to positive depression detection	
The Third Eye: Vintel Platform <i>Multi-Agent LLM Cybersecurity Intelligence System</i>	Jan 2026 – May 2026
• Designed a multi-agent LLM pipeline with two-round editorial deliberation and Borda-count cross-story ranking algorithm	
• Built persistent memory and A2A protocol for autonomous agent coordination across polyglot microservices (Go, TS, Rust)	
HeadBall2 Game Player <i>Self Project</i> Real-Time Game Automation with CV and RL	Feb 2025 – Mar 2025
• Developed a real-time computer vision pipeline for mobile game automation using OpenCV , from live video streams at 30 FPS	
• Designed a reinforcement learning system with WebSocket-based client-server communication, achieving 50 ms end-to-end latency	

Research & Publications

Parameter Specific Literature Retrieval <i>Master’s Thesis Project (MTP)</i> Prof. Balamurugan Palaniappan	Jul 2025 – Nov 2025
• Streamlined a biomedical literature retrieval pipeline to search PubMed data and extract parameter-specific clinical evidence	
• Improved entity extraction F1-score to 90% by fine-tuning a PubMedBERT-based NER model on a manually annotated dataset	
• Converted unstructured NER output into structured XML using a BART-based Seq2Tree model for parameter range extraction	
• Implemented a distributed ML pipeline with Ray, PyTorch, and HuggingFace Transformers ; deployed it with Next.js/FastAPI	
• Built a provenance-tracked Neo4j knowledge graph and QA engine reaching an 88% answer rate with every claim cited	
Classification of Plant Diseases using SOTA Deep learning Algorithms	July 2024
<i>International Journal of Computing</i> DOI: 10.47839/ijc.23.4.3542	
• Implemented and evaluated multiple CNN architectures on a dataset of 87,900+ leaf images across 38 classes	
• Achieved up to 97.31% validation accuracy , showing robust generalization across multiple plant diseases	
Multimodal Transformer Architecture and Optimization <i>Seminar</i> Prof. Urban Larsson	Jan 2025 – Apr 2025
• Implemented the Attention is All You Need paper in PyTorch and presented an in-depth analysis of the paper during the seminar	
• Analyzed in-depth concepts of An Image is Worth 16x16 Words and presented how to use images in transformer architecture	
• Conducted a seminar presentation on the research behind the LoRA paper and discussed its advantages and disadvantages	

Position of Responsibility

Teaching Assistant <i>IE 201, AI & ML Lab, IIT Bombay</i>	Jan 2026 – Apr 2026
• Designed lab experiments , set and graded exams for 40 B.Tech students , and conducted doubt sessions on AI/ML concepts	
Teaching Assistant <i>IE 643, Deep Learning – Theory and Practice</i>	Jul 2025 – Nov 2025
• Led weekly programming sessions for 80+ students ; managed grading, project planning, and regular doubt sessions	

Achievements

• Published research paper on Plant Disease Classification using Deep Learning in International Journal of Computing	
• Secured World Rank 32 in CodeChef Starters 42 competitive programming contest	
• Secured Rank 1 in HackForCG Hackathon by developing an ML-based computer vision solution using OpenCV and Haar Cascade	